EDA for the road features extracted from contextual database

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2026-09-09

EDA Report: Dataset (model_dataset.csv)

Introduction

This report presents an exploratory data analysis (EDA) of `model_dataset.csv`, a dataset of 49 road segments in Western Australia. The contextual variables (e.g. population statistics, distance to public transport, health, and education facilities, building and retail counts) were extracted from a contextual ecosocial database, while the image-derived variables (e.g. footpath type, land use, and the presence of nearby amenities) were obtained from earlier feature extraction work. The goal is to identify variables that are associated with observed pedestrian counts, in preparation for building a predictive regression model.

Dataset and Variables

The dataset contains two response variables, measuring pedestrian counts per hour in two directions:

- `obs_peak_along_ped_per_h`
- `obs_peak_cross_ped_per_h`

The remaining variables fall into two groups: 21 numeric variables (e.g. building counts, population density, distance to public transport, health, and education facilities) and 16 categorical variables (e.g. footpath type, land use, and the presence of nearby amenities).

```
road_features <- read.csv("data/model_dataset.csv", stringsAsFactors = FALSE)

comma_numeric_cols <- c(
  "abs_mb_dwellings_per_sqkm_2021", "abs_mb_persons_per_sqkm_2021",
  "pta_nearest_active_stop_distance_m", "health_nearest_establishment_distance_m",
  "education_nearest_active_school_distance_m", "ms_building_footprint_area_m2_500m"
)
for (col in comma_numeric_cols) {
  road_features[[col]] <- as.numeric(gsub(",", "", road_features[[col]]))
}

response_vars <- c("obs_peak_cross_ped_per_h", "obs_peak_along_ped_per_h")

numeric_vars <- c(
  "abs_mb_area_sqkm_2021", "abs_mb_dwellings_2021", "abs_mb_persons_usual_resident_2021",
  "abs_mb_dwellings_per_sqkm_2021", "abs_mb_persons_per_sqkm_2021",
  "pta_nearest_active_stop_distance_m", "pta_active_stops_within_500m",
  "health_nearest_establishment_distance_m",
  "education_nearest_active_school_distance_m",
  "ms_building_count_500m", "ms_building_count_per_sqkm_500m",
  "ms_building_footprint_area_m2_500m", "ms_building_coverage_ratio_500m",
  "google_food_drink_count_500m", "google_grocery_convenience_market_count_500m",
```

```
  "google_other_retail_count_500m",
  "pedestrian_crossings", "commercial_buildings", "residential_buildings",
  "industrial_buildings", "parking_areas"
)

categorical_vars <- c(
  "abs_mb_category_2021", "pta_nearest_active_stop_type",
  "health_nearest_establishment_category", "health_nearest_establishment_type",
  "education_nearest_school_sector", "education_nearest_school_low_year",
  "education_nearest_school_high_year",
  "google_major_shopping_present_500m",
  "footpath_both_sides", "footpath_one_side",
  "fencing_barriers", "railway_station", "bus_stops", "overpass_bridge",
  "park_bushland_reserve", "shared_pedestrian_and_cycle_path"
)

analysis_df <- road_features[, c(response_vars, numeric_vars, categorical_vars)]
```

No missing values were found in any of the variables used in this analysis.

Numeric Variables

Correlation coefficients were calculated between each numeric variable and both response variables.

```
cor_matrix <- sapply(response_vars, function(t) {
  sapply(numeric_vars, function(v) cor(analysis_df[[v]], analysis_df[[t]]))
})
round(cor_matrix, 2)
```

```
##                                                obs_peak_cross_ped_per_h
## abs_mb_area_sqkm_2021                                             -0.15
## abs_mb_dwellings_2021                                             -0.05
## abs_mb_persons_usual_resident_2021                                 0.05
## abs_mb_dwellings_per_sqkm_2021                                    -0.12
## abs_mb_persons_per_sqkm_2021                                      -0.01
## pta_nearest_active_stop_distance_m                                 0.16
## pta_active_stops_within_500m                                      -0.06
## health_nearest_establishment_distance_m                            0.00
## education_nearest_active_school_distance_m                        -0.15
## ms_building_count_500m                                             0.08
## ms_building_count_per_sqkm_500m                                    0.08
## ms_building_footprint_area_m2_500m                                 0.10
## ms_building_coverage_ratio_500m                                    0.10
## google_food_drink_count_500m                                       0.40
## google_grocery_convenience_market_count_500m                       0.50
## google_other_retail_count_500m                                     0.44
## pedestrian_crossings                                               0.23
## commercial_buildings                                               0.66
## residential_buildings                                             -0.21
## industrial_buildings                                              -0.08
## parking_areas                                                      0.42
##                                                obs_peak_along_ped_per_h
## abs_mb_area_sqkm_2021                                             -0.13
## abs_mb_dwellings_2021                                             -0.07
## abs_mb_persons_usual_resident_2021                                -0.03
```

```
## abs_mb_dwellings_per_sqkm_2021                          -0.09
## abs_mb_persons_per_sqkm_2021                            -0.05
## pta_nearest_active_stop_distance_m                      -0.08
## pta_active_stops_within_500m                             0.02
## health_nearest_establishment_distance_m                 -0.01
## education_nearest_active_school_distance_m              -0.10
## ms_building_count_500m                                   0.09
## ms_building_count_per_sqkm_500m                          0.09
## ms_building_footprint_area_m2_500m                       0.12
## ms_building_coverage_ratio_500m                          0.12
## google_food_drink_count_500m                             0.46
## google_grocery_convenience_market_count_500m             0.52
## google_other_retail_count_500m                           0.55
## pedestrian_crossings                                     0.18
## commercial_buildings                                     0.77
## residential_buildings                                   -0.14
## industrial_buildings                                    -0.08
## parking_areas                                            0.46
```

Findings:

Five variables show a clear correlation ($|r| >= 0.4$) with both response variables: `commercial_buildings`, `google_other_retail_count_500m`, `google_grocery_convenience_market_count_500m`, `parking_areas`, and `google_food_drink_count_500m`. All five are statistically significant ($p < 0.01$), with correlation coefficients ranging from 0.40 to 0.77. These variables all relate to commercial and retail activity near the site.

Two variables, `pedestrian_crossings` and `residential_buildings`, show weak correlations ($|r|$ between 0.14 and 0.23) that are not statistically significant.

The remaining variables, including ABS population statistics, distance to public transport, health, and education facilities, and Microsoft building footprint measures, show little to no correlation ($|r| < 0.2$) with either target.

```
selected_vars <- c(
  "commercial_buildings", "google_other_retail_count_500m",
  "google_grocery_convenience_market_count_500m", "parking_areas",
  "google_food_drink_count_500m", "pedestrian_crossings", "residential_buildings"
)

library(PerformanceAnalytics)
```

```
## Loading required package: xts
```

```
## Loading required package: zoo
```

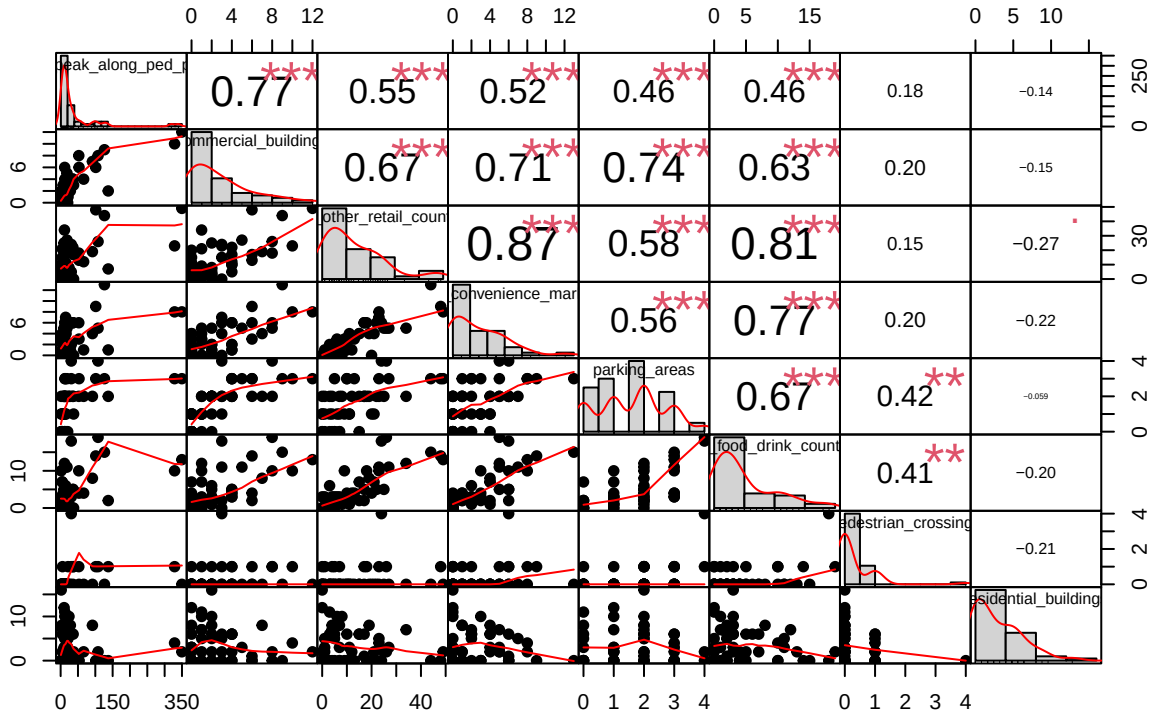
```
##
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

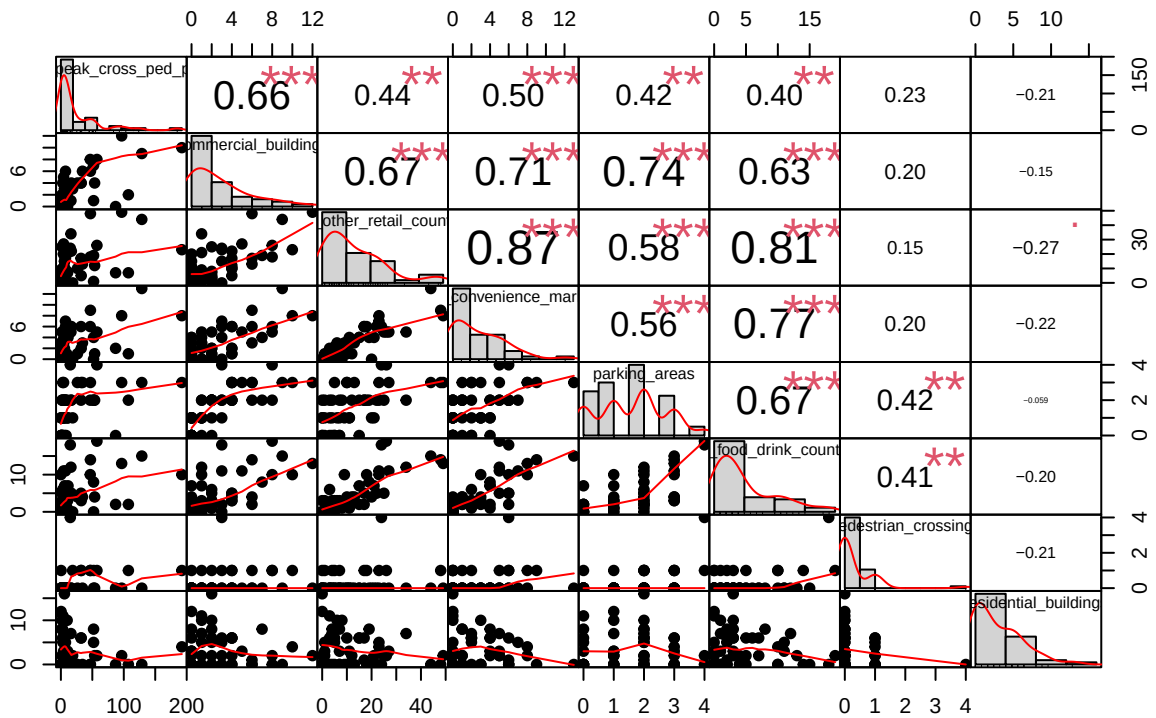
```
##
## Attaching package: 'PerformanceAnalytics'
```

```
## The following object is masked from 'package:graphics':
##
##     legend
```

```
chart.Correlation(analysis_df[, c("obs_peak_along_ped_per_h", selected_vars)], histogram = TRUE, pch =
```




```
chart.Correlation(analysis_df[, c("obs_peak_cross_ped_per_h", selected_vars)], histogram = TRUE, pch =
```



The five significant predictors are also highly correlated with each other (r = 0.56 to 0.87). This indicates strong multicollinearity, which will need to be addressed before these variables can be used together in a regression model.

Logical Categorical Variables

The number of TRUE and FALSE cases was checked for each logical variable, to identify variables with insufficient sample size in either group.

```
logical_cat_vars <- categorical_vars[sapply(analysis_df[categorical_vars], is.logical)]

count_table <- data.frame(
  variable = logical_cat_vars,
  FALSE_n  = sapply(logical_cat_vars, function(v) sum(analysis_df[[v]] == FALSE, na.rm = TRUE)),
  TRUE_n   = sapply(logical_cat_vars, function(v) sum(analysis_df[[v]] == TRUE, na.rm = TRUE))
)
knitr::kable(count_table, caption = "TRUE/FALSE counts for logical variables (n=49)")
```

Table 1: TRUE/FALSE counts for logical variables (n=49)

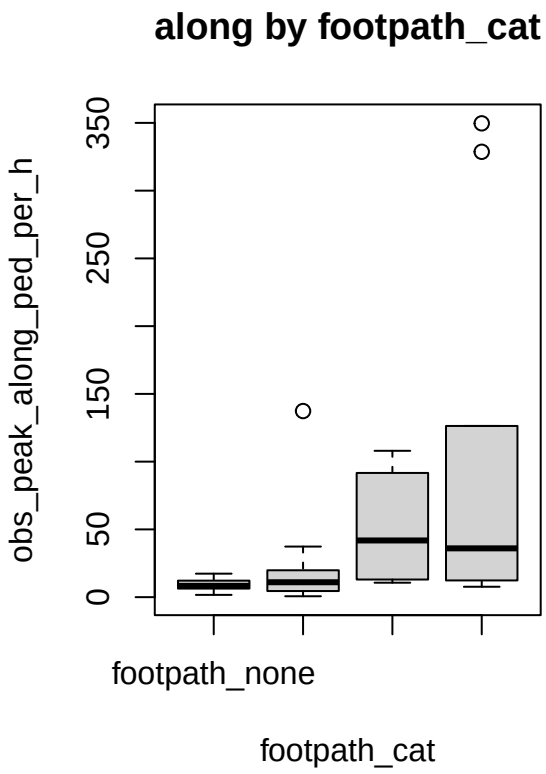
	variable	FALSE_n	TRUE_n
google_major_shopping_present_500m	google_major_shopping_present_500m	34	15
footpath_both_sides	footpath_both_sides	30	19
footpath_one_side	footpath_one_side	16	33
fencing_barriers	fencing_barriers	3	46
railway_station	railway_station	47	2
bus_stops	bus_stops	47	2
overpass_bridge	overpass_bridge	48	1
park_bushland_reserve	park_bushland_reserve	8	41
shared_pedestrian_and_cycle_path	shared_pedestrian_and_cycle_path	36	13

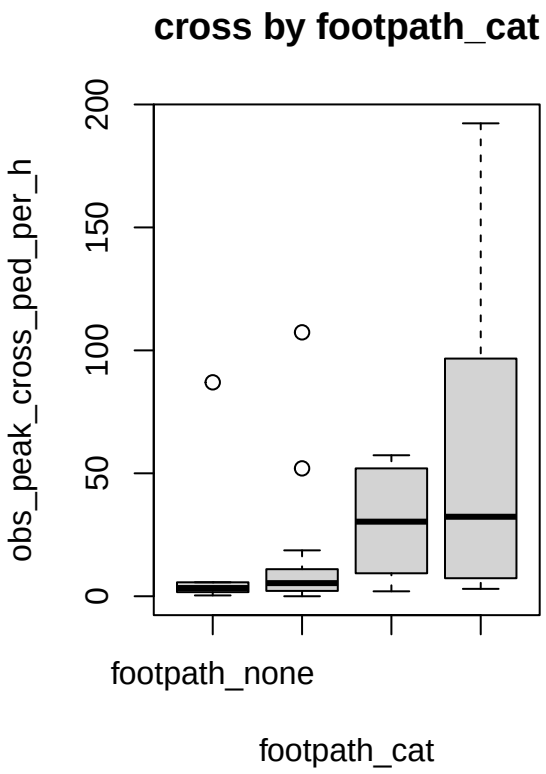
Four variables have fewer than 5 cases in one group — `fencing_barriers`, `railway_station`, `bus_stops`, and `overpass_bridge` — and are excluded from further analysis, as the small group size makes comparisons unreliable.

The remaining variables are examined with boxplots. `footpath_both_sides` and `footpath_one_side` are combined into a single four-level variable, `footpath_cat` (`footpath_none`, `footpath_one_side`, `footpath_mixed`, `footpath_both_sides`), since a segment can have footpaths on one side, both sides, or a mix of the two.

```
analysis_df$footpath_cat <- factor(
  ifelse(analysis_df$footpath_both_sides & analysis_df$footpath_one_side, "footpath_mixed",
  ifelse(analysis_df$footpath_both_sides, "footpath_both_sides",
  ifelse(analysis_df$footpath_one_side, "footpath_one_side", "footpath_none"))),
  levels = c("footpath_none", "footpath_one_side", "footpath_mixed", "footpath_both_sides")
)

par(mfrow = c(1, 2))
boxplot(analysis_df$obs_peak_along_ped_per_h ~ analysis_df$footpath_cat,
        xlab = "footpath_cat", ylab = "obs_peak_along_ped_per_h", main = "along by footpath_cat")
boxplot(analysis_df$obs_peak_cross_ped_per_h ~ analysis_df$footpath_cat,
        xlab = "footpath_cat", ylab = "obs_peak_cross_ped_per_h", main = "cross by footpath_cat")
```

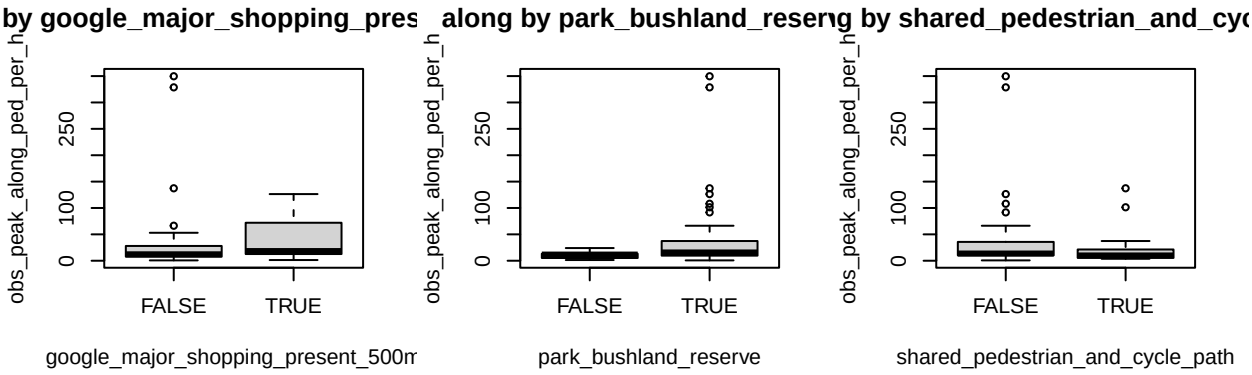


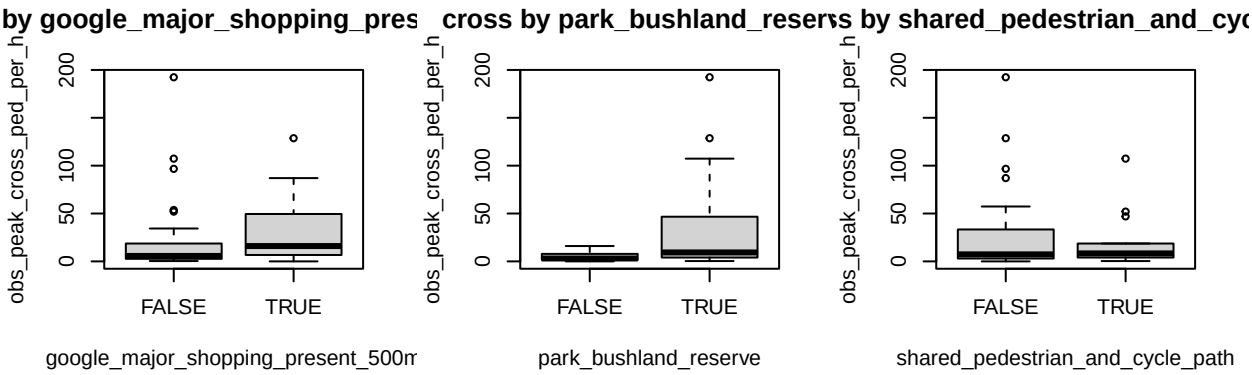



```
par(mfrow = c(1, 1))
```

```
kept_logical_vars <- c("google_major_shopping_present_500m", "park_bushland_reserve", "shared_pedestrian

par(mfrow = c(2, 3))
for (v in kept_logical_vars) {
  boxplot(analysis_df$obs_peak_along_ped_per_h ~ analysis_df[[v]],
          xlab = v, ylab = "obs_peak_along_ped_per_h", main = paste("along by", v))
}
for (v in kept_logical_vars) {
  boxplot(analysis_df$obs_peak_cross_ped_per_h ~ analysis_df[[v]],
          xlab = v, ylab = "obs_peak_cross_ped_per_h", main = paste("cross by", v))
}
```






```
par(mfrow = c(1, 1))
```

Findings:

For `footpath_cat`, pedestrian counts increase steadily across the four categories for both response variables. `footpath_none` and `footpath_one_side` show low medians and narrow spread, while `footpath_mixed` and `footpath_both_sides` show noticeably higher medians and wider spread, including several high outliers. This suggests that more extensive footpath coverage is associated with higher pedestrian counts.

`google_major_shopping_present_500m` and `park_bushland_reserve` show a similar pattern: locations where these are TRUE have a higher median and wider spread than locations where they are FALSE, for both target variables.

`shared_pedestrian_and_cycle_path` shows little difference between the two groups, with overlapping boxes and similar medians.

Character-Type Categorical Variables

The level distribution of each character-type categorical variable was checked.

```
char_cat_vars <- categorical_vars[sapply(analysis_df[categorical_vars], is.character)]

level_table <- do.call(rbind, lapply(char_cat_vars, function(v) {
  tab <- table(analysis_df[[v]])
  data.frame(variable = v, level = names(tab), n = as.integer(tab),
             pct = round(100 * as.integer(tab) / sum(tab), 1))
}))
rownames(level_table) <- NULL
knitr::kable(level_table, caption = "Level counts and percentages for character categorical variables")
```

Table 2: Level counts and percentages for character categorical variables

variable	level	n	pct
abs_mb_category_2021	Commercial	10	20.4
abs_mb_category_2021	Education	3	6.1
abs_mb_category_2021	Industrial	3	6.1
abs_mb_category_2021	Parkland	4	8.2
abs_mb_category_2021	Primary Production	1	2.0
abs_mb_category_2021	Residential	26	53.1
abs_mb_category_2021	Transport	2	4.1
pta_nearest_active_stop_type	School	10	20.4
pta_nearest_active_stop_type	Standard	11	22.4
pta_nearest_active_stop_type	Transwa	28	57.1
health_nearest_establishment_category	Acute Hospital	44	89.8
health_nearest_establishment_category	Nursing Post	5	10.2
health_nearest_establishment_type	Private	4	8.2
health_nearest_establishment_type	Public	45	91.8
education_nearest_school_sector	GOVERNMENT	31	63.3
education_nearest_school_sector	NON-GOVERNMENT	18	36.7
education_nearest_school_low_year	KIN	35	71.4
education_nearest_school_low_year	PKG	8	16.3
education_nearest_school_low_year	Y07	4	8.2
education_nearest_school_low_year	Y10	1	2.0
education_nearest_school_low_year	Y11	1	2.0
education_nearest_school_high_year	Y06	32	65.3
education_nearest_school_high_year	Y10	4	8.2
education_nearest_school_high_year	Y12	13	26.5

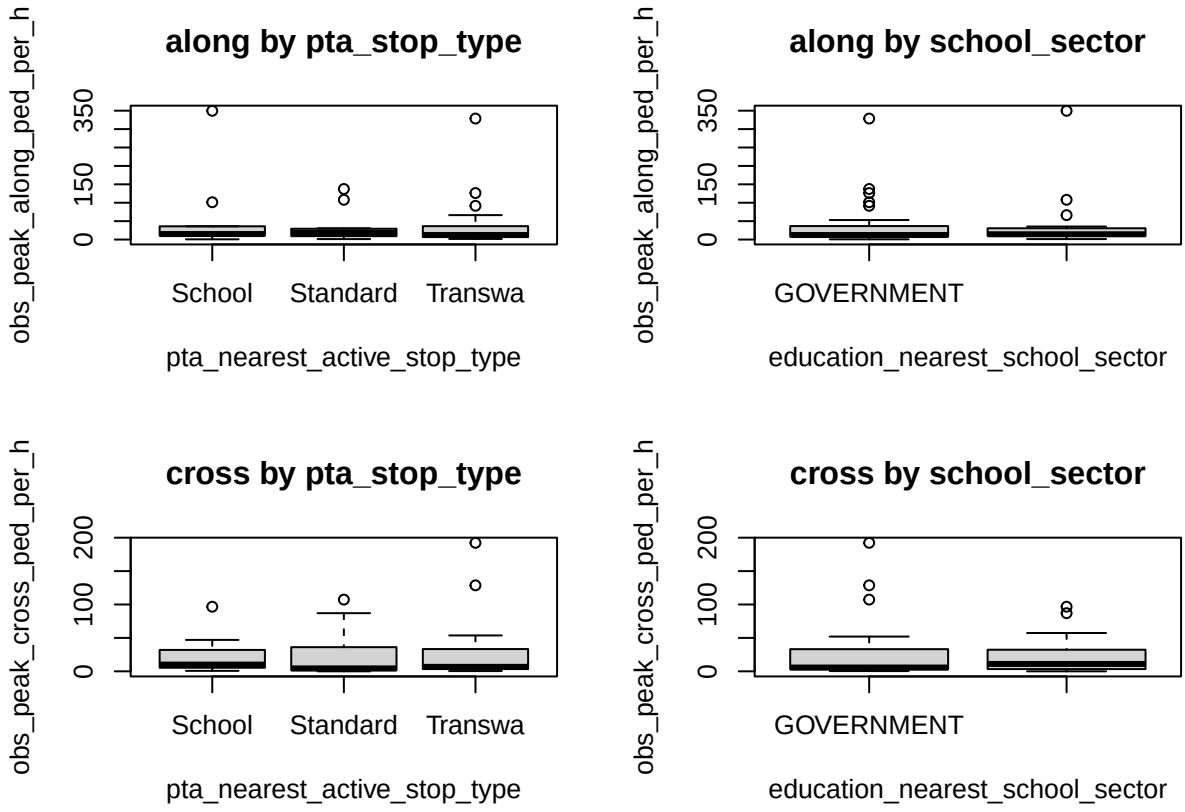
Findings:

Two variables have a reasonable balance across all levels: `pta_nearest_active_stop_type` (School=10, Standard=11, Transwa=28) and `education_nearest_school_sector` (GOVERNMENT=31, NON-GOVERNMENT=18).

`abs_mb_category_2021` is dominated by Residential (53.1%) and Commercial (20.4%), with the remaining five levels each below 10% of the sample.

The other four variables have at least one level with fewer than 5 observations — `health_nearest_establishment_category`, `health_nearest_establishment_type`, `education_nearest_school_low_year`, and `education_nearest_school_high_ye` — and are excluded from further analysis for the same reason as above.

```
par(mfrow = c(2, 2))
boxplot(analysis_df$obs_peak_along_ped_per_h ~ analysis_df$pta_nearest_active_stop_type,
        xlab = "pta_nearest_active_stop_type", ylab = "obs_peak_along_ped_per_h", main = "along by pta_
boxplot(analysis_df$obs_peak_along_ped_per_h ~ analysis_df$education_nearest_school_sector,
        xlab = "education_nearest_school_sector", ylab = "obs_peak_along_ped_per_h", main = "along by s
boxplot(analysis_df$obs_peak_cross_ped_per_h ~ analysis_df$pta_nearest_active_stop_type,
        xlab = "pta_nearest_active_stop_type", ylab = "obs_peak_cross_ped_per_h", main = "cross by pta_
boxplot(analysis_df$obs_peak_cross_ped_per_h ~ analysis_df$education_nearest_school_sector,
        xlab = "education_nearest_school_sector", ylab = "obs_peak_cross_ped_per_h", main = "cross by s
```



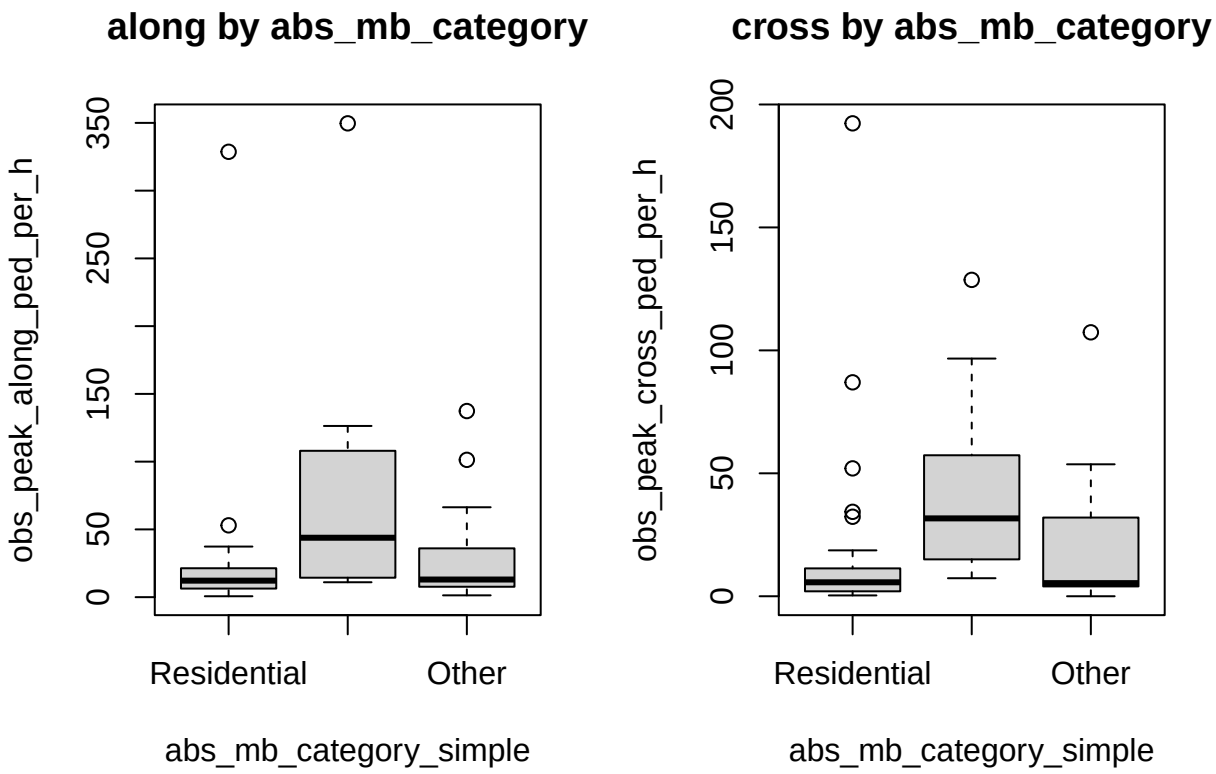

```
par(mfrow = c(1, 1))
```

Neither `pta_nearest_active_stop_type` nor `education_nearest_school_sector` shows a meaningful difference in pedestrian counts across groups, based on these plots.

`abs_mb_category_2021` is simplified into three levels — `Residential`, `Commercial`, and `Other` (combining Education, Industrial, Parkland, Primary Production, and Transport) — since the smaller levels each account for less than 10% of the sample.

```
analysis_df$abs_mb_category_simple <- factor(
  ifelse(analysis_df$abs_mb_category_2021 %in% c("Residential", "Commercial"),
         analysis_df$abs_mb_category_2021, "Other"),
  levels = c("Residential", "Commercial", "Other")
)

par(mfrow = c(1, 2))
boxplot(analysis_df$obs_peak_along_ped_per_h ~ analysis_df$abs_mb_category_simple,
        xlab = "abs_mb_category_simple", ylab = "obs_peak_along_ped_per_h", main = "along by abs_mb_cat
boxplot(analysis_df$obs_peak_cross_ped_per_h ~ analysis_df$abs_mb_category_simple,
        xlab = "abs_mb_category_simple", ylab = "obs_peak_cross_ped_per_h", main = "cross by abs_mb_cat
```




```
par(mfrow = c(1, 1))
```

The Commercial group shows the highest median and widest spread, followed by Other, with Residential showing the lowest and narrowest distribution, for both target variables. This is consistent with the earlier finding that `commercial_buildings` was the strongest numeric predictor.

Conclusion

`commercial_buildings` and related retail/commercial density measures (`google_other_retail_count_500m`, `google_grocery_convenience_market_count_500m`, `parking_areas`, `google_food_drink_count_500m`) are the strongest and most consistent numeric predictors of pedestrian counts, but they are highly correlated with each other and require a multicollinearity check before joint use in a model.

Among categorical variables, `footpath_cat`, `google_major_shopping_present_500m`, `park_bushland_reserve`, and land-use category (`abs_mb_category_simple`) show a clear relationship with pedestrian counts, all pointing toward higher counts in commercially developed, well-footpathed areas. `shared_pedestrian_and_cycle_path`, `pta_nearest_active_stop_type`, and `education_nearest_school_sector` show no meaningful relationship and are unlikely to add value to the model.

Recommended next step: examine multicollinearity among the commercial/retail density variables (e.g. via VIF or correlation-based selection) before fitting a regression model, and confirm whether `footpath_cat` and `abs_mb_category_simple` add explanatory power independent of `commercial_buildings`.